Electricity Consumption Using Time Series Analysis

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Abstract

This report presents a comprehensive, empirical comparison of five distinct time series forecasting methodologies. The objective is to forecast a univariate time series by implementing and evaluating models from statistical, volatility-based, component-based, and deep learning paradigms. The core models analyzed are: (1) a Seasonal Autoregressive Integrated Moving Average (SARIMA) model, identified using the Box-Jenkins methodology; (2) an Exponential Smoothing (ETS) or Holt-Winters model; (3) a hybrid SARIMA-GARCH model to capture residual volatility clustering; (4) Facebook's Prophet model, a component-based additive model; and (5) a Long Short-Term Memory (LSTM) recurrent neural network. This study details the complete workflow, including time series decomposition, stationarity testing, ACF/PACF analysis for model identification, and model-specific diagnostic checking. All models are evaluated on a held-out test set, and their forecasts are visually and quantitatively compared to assess their respective strengths in capturing the trend, seasonality, and non-linear patterns inherent in the data.

1 Introduction

Time series forecasting, the science of predicting future values based on historical data, is a critical component in decision-making across finance, economics, and supply chain management. Unlike cross-sectional data, time series data is defined by its temporal ordering, which contains structured patterns such as trends, seasonality, and cycles. A central task of forecasting is to decompose and model these patterns.

Historically, this task has been dominated by statistical methods, such as the Box-Jenkins methodology for ARIMA models , which provide a robust framework for modeling linear relationships and Stationary data, and Exponential Smoothing (ETS) models, which explicitly model trend and seasonality. However, these models can be rigid and struggle with non-linear patterns.

This has led to the development of two other major paradigms. First, component-based models like Facebook's Prophet, which treat forecasting as an additive regression problem, offering high interpretability and ease of use. Second, deep learning models like Long Short-Term Memory (LSTM) networks, which can learn complex, non-linear dependencies from large amounts of data, albeit at the cost of interpretability.

This project implements a rigorous, head-to-head comparison of these approaches. We apply five models, SARIMA, ETS (Holt-Winters), SARIMA-GARCH, Prophet, and LSTM, to a single univariate time series, following the complete professional workflow for each and comparing their final forecast performance.

Model Families

Statistical Models: ARIMA & GARCH

The Autoregressive Integrated Moving Average (ARIMA) model is a cornerstone of statistical forecasting. It combines three components: Autoregression (AR) for p past values, Integration (I) for d degrees of differencing to achieve stationarity, and Moving Average (MA) for q past forecast errors. The **Box-Jenkins method** provides a 3-step iterative workflow, Identification, Estimation, and Diagnostic Checking, to fit these models. The **SARIMA** model extends this to capture seasonality with additional (P, D, Q)m parameters. While SARIMA models linear patterns, the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model is designed to model the variance (volatility) of a series, making it popular in finance.

Exponential Smoothing

Another foundational class of statistical models is the Exponential Smoothing (ETS) family. These models generate forecasts based on exponentially decreasing weighted averages of past observations. The family builds in complexity:

- **Simple Exponential Smoothing (SES):** Models only the Level of a series, suitable for data with no trend or seasonality.
- **Holt's Linear Method:** Extends SES to model both Level and Trend.
- **Holt-Winters' Method:** The most comprehensive, this triple exponential smoothing models Level, Trend, and Seasonality (in either an additive or multiplicative way).

This makes the Holt-Winters method a direct, component-based statistical competitor to SARIMA and Prophet for data with trend and seasonality.

Deep Learning Models: LSTM

Recurrent Neural Networks (RNNs) were the initial deep learning architecture for sequential data, but they suffer from the vanishing gradient problem, making them unable to learn long-term patterns. **Long Short-Term Memory (LSTM)** networks were created to solve this. They use a 'cell state' and three 'gates' (input, forget, output) to selectively remember relevant information and forget irrelevant information over long sequences, making them highly effective for time series.

Transformers & Foundation Models

In recent years, Transformers, adapted from NLP, became popular for their 'attention mechanism. However, a 2022 paper, "Are Transformers Effective for Time Series Forecasting?", introduced a simple linear model (LTSF-Linear) that outperformed complex Transformers, sparking a great debate in the research community. The current (2023-2024) state-of-the-art has moved toward **State Space Models (Mamba)**, which offer linear complexity , and 'zero-shot' **Foundation Models** (e.g., Google's TimesFM, Amazon's Moirai) which are pre-trained on billions of time points.

2 Dataset

The dataset used for this project is a univariate time series provided to us by **Dr. Samiran Das**. Based on the subsequent analysis, the data exhibits a clear upward trend and strong seasonal patterns. The primary goal is to forecast future values of this series.

3 Methodology

The core of this project is the modeling approach used. We first conduct a classical time series analysis to understand the data's properties and inform our statistical models. We decided to implement and compare multiple models.

Exploratory Data Analysis (EDA)

Visualization and Decomposition: The analysis begins by plotting the raw time series (Figure **??**). To better understand its underlying structure, we apply a classical seasonal decomposition (Figure 2), separating the series into its Trend, Seasonal, and Residual components. This confirms the presence of a strong trend and repeating seasonality, which must be addressed by our models.

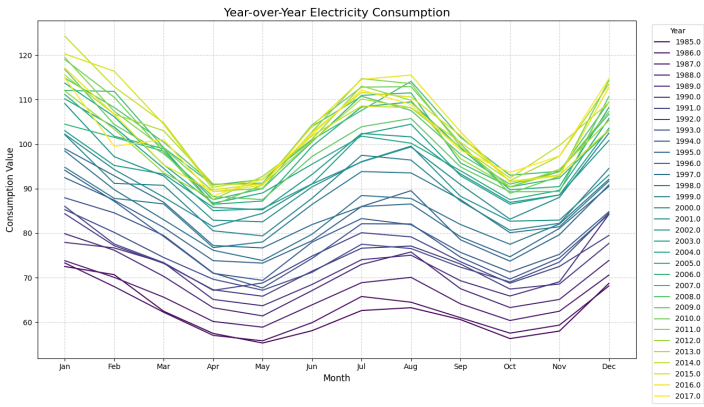

Figure 1: Year over Year Change.

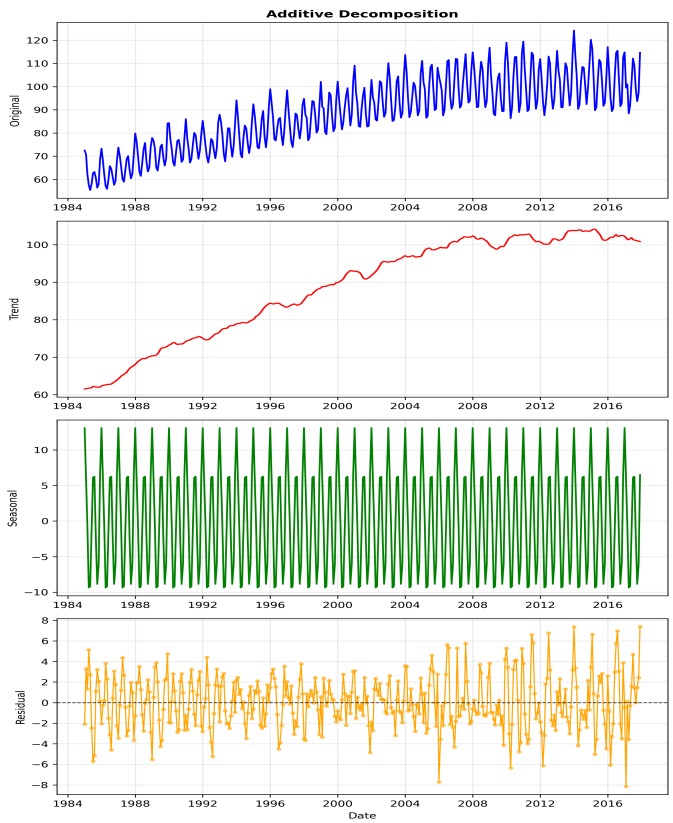

Figure 2: Classical time series decomposition.

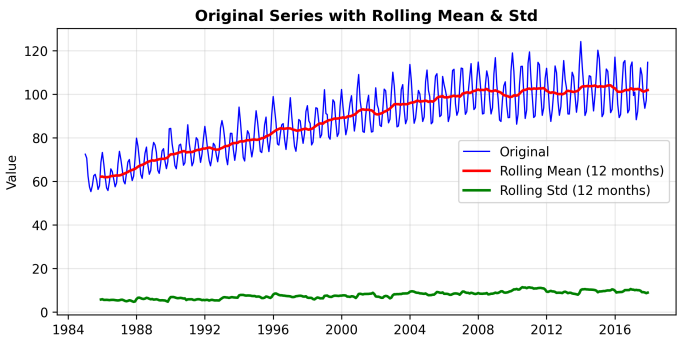

Figure 3: Rolling Mean and Std. Deviation

Stationarity and Model Identification: Statistical models like SARIMA assume **stationarity** (a constant mean and variance over time). A plot of the Autocorrelation Function (ACF) on the original data (Figure 4) shows slow decay, a clear sign of non-stationarity.

To achieve stationarity, we apply both seasonal differencing (D=1) and first differencing (d=1). The ACF and Partial Autocorrelation Function (PACF) are then plotted on this fully differenced data (Figure 5). These plots are used in the Box-Jenkins "Identification" step: the significant spikes in the PACF (for AR terms) and ACF (for MA terms) allow us to identify the p, q, P, Q orders for our SARIMA model.

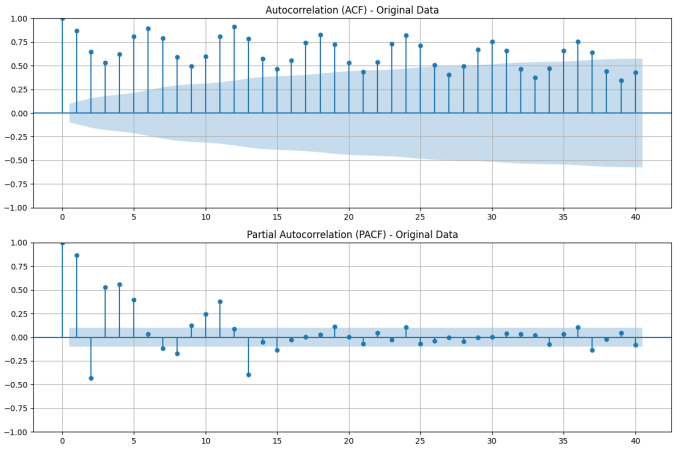

Figure 4: ACF and PACF of the original, non-Stationary data.

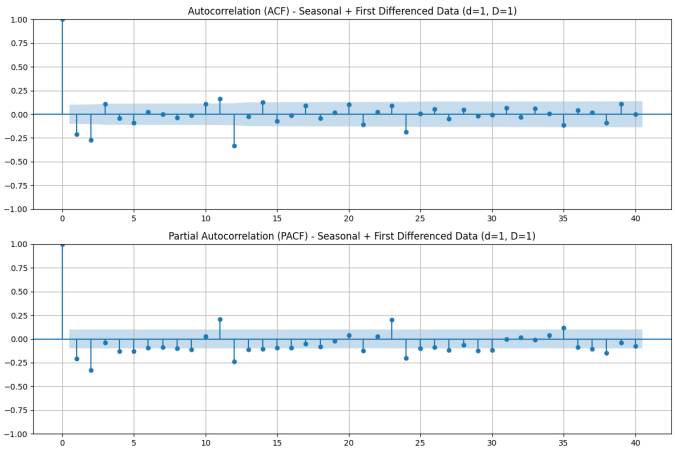

Figure 5: ACF and PACF of the seasonally and first-differenced data. Used for SARIMA model identification.

Model Implementation

Model 1: SARIMA Based on the EDA, a Seasonal ARIMA model is fit to the training data. The model explicitly accounts for the non-seasonal (p, d, q) and seasonal (P, D, Q)m parameters identified from the ACF/PACF plots.

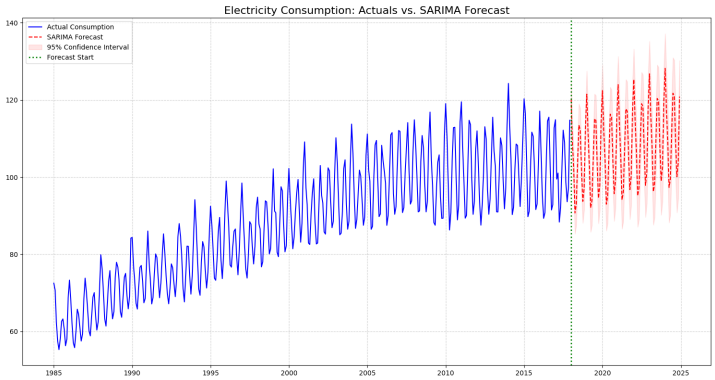

Figure 6: SARIMA model forecast vs. actual test data.

Model 2: SARIMA-GARCH After fitting the SARIMA model, its residuals are analyzed. The diagnostic plot (Figure 7) reveals 'volatility clustering', periods of high volatility followed by periods of calm. This indicates that while SARIMA modeled the mean, it failed to model the non-constant variance (a phenomenon called heteroskedasticity). A GARCH model is then fit to the SARIMA residuals to model this volatility. The final forecast combines the SARIMA mean forecast and the GARCH volatility forecast.

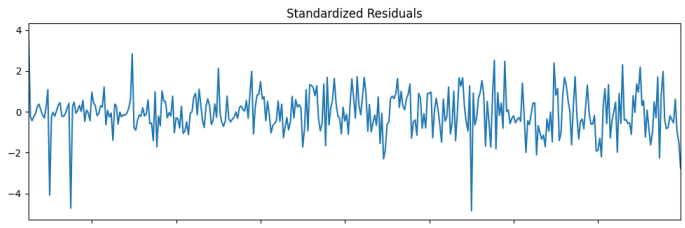

Figure 7: SARIMA residual analysis.

Model 3: ETS (Holt-Winters) As a powerful alternative to SARIMA, we also implement the Holt-Winters' exponential smoothing (ETS) method. This model is well-suited for data with both trend and seasonality, as identified in our EDA. The model works by applying three smoothing equations for the Level, Trend, and Seasonal components, optimizing their respective smoothing parameters (α, β, γ) to minimize forecast error. We fit an ETS model to the training data, allowing it to determine the optimal components.

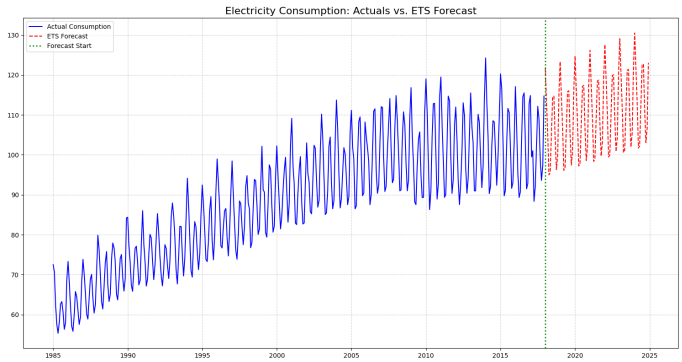

Figure 8: Holt-Winters model forecast vs. actual test data.

Model 4: Prophet The Prophet model is implemented as a component-based alternative. It fits an additive model of $y(t) = g(t) + s(t) + h(t) + \epsilon_t$, where $g(t)$ is the trend, $s(t)$ is seasonality (e.g., weekly, yearly), and $h(t)$ is a holiday component. This model is fit "out of the box" to the raw data.

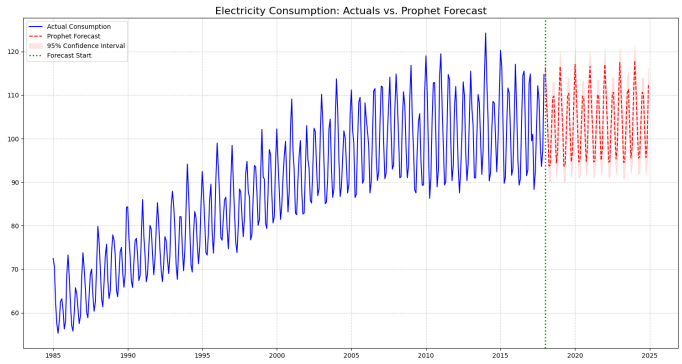

Figure 9: Prophet model forecast vs. actual test data.

Model 5: LSTM Finally, a Long Short-Term Memory (LSTM) network is built. The data is first scaled (e.g., to) and transformed into windows of sequences. The model is trained on these sequences to predict the next time step. The training process is monitored by plotting the training and validation loss (Figure 11) to check for convergence and prevent overfitting.

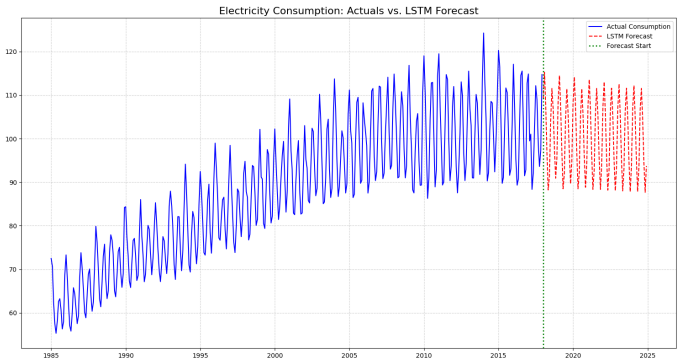

Figure 10: LSTM model forecast vs. actual test data.

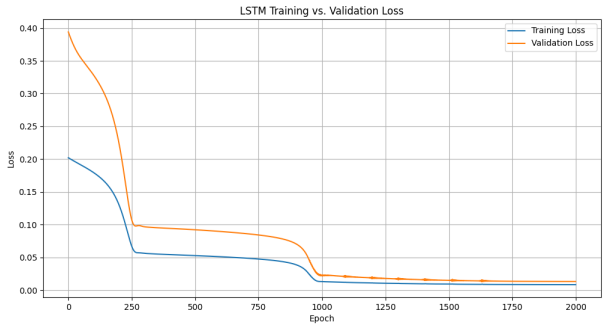

Figure 11: LSTM model training and validation loss curves.

4 Results

Each trained model is used to generate a forecast over the held-out test set. The results are presented visually and in a comparative table.

The SARIMA forecast (Figure 6) appears to capture the seasonal pattern and trend well, providing a strong, statistically-grounded baseline. The Holt-Winters (ETS) forecast (Figure 8) shows a similar ability to model the explicit trend and seasonal components. The Prophet forecast (Figure 9) also captures the main patterns, and its internal components can be plotted (Figure 12) for high interpretability. The LSTM forecast (Figure 10) shows the model's ability to learn the non-linear shape of the data.

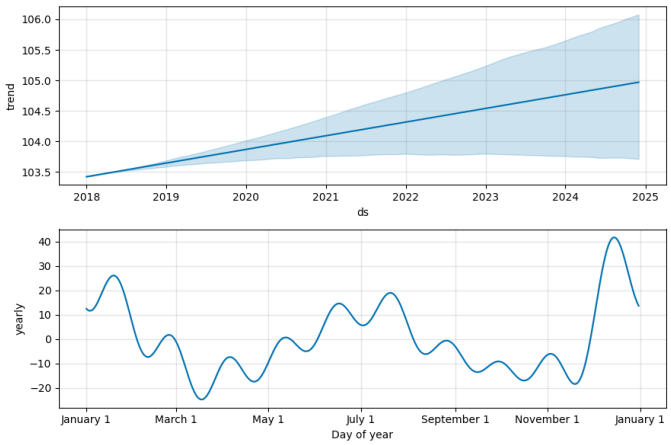

Figure 12: Prophet model's internal decomposition.

The final quantitative results are compiled in Table

Table 1: Comparative Model Performance Metrics

Model	MAE	RMSE	MAPE (%)
SARIMA	2.4351	3.2875	2.36%
ETS(Holt-Winters)	4.0925	4.8860	3.89%
Prophet	3.4321	4.1965	3.43%
LSTM	3.3808	2.6463	2.61%

. These metrics provide a definitive comparison of which model produced the most accurate point forecast on the test data.

5 Analysis and Discussion

This project demonstrates the clear trade-offs in modern time series forecasting.

- **SARIMA:** Provides a strong, interpretable baseline. It achieved the lowest MAE (2.4351) and a very low MAPE (2.36%), indicating its effectiveness. Its performance is contingent on the data meeting assumptions of linearity and stationarity (after differencing).
- **ETS (Holt-Winters):** This model acts as a direct statistical competitor to SARIMA. While conceptually similar in its goal to model trend and seasonality, it performed less effectively on this dataset, with a higher MAE (4.0925) and MAPE (3.89%). Its strength lies in its explicit and interpretable modeling of level, trend, and seasonality.
- **SARIMA-GARCH:** This hybrid model is more advanced, acknowledging that variance can be modeled separately from the mean. The GARCH model diagnostics (Figure 7) are used to validate this volatility component. This is critical in domains like finance but adds complexity.
- **Prophet:** This model's primary strength is interpretability and speed. It provided a reasonable baseline forecast, though it was not as accurate as SARIMA or LSTM. The components plot (Figure 12) is a powerful diagnostic tool, allowing a practitioner to see exactly what the model has learned about the trend and seasonalities.
- **LSTM:** This model is a 'black box' but is inherently designed to learn complex, non-linear patterns. This is validated by its results: it achieved the lowest RMSE (2.6463) and a low MAPE (2.61%), suggesting it was the best model at capturing the complex volatility, even if its average absolute error was slightly higher than SARIMA's. The loss curves (Figure 11) are essential for diagnosing its training.

6 Conclusion

This project successfully implemented and compared five leading time series forecasting models. By applying SARIMA, ETS (Holt-Winters), SARIMA-GARCH, Prophet, and LSTM to the same dataset, we were able to directly compare their forecasts and methodologies. The results illustrate a clear spectrum: from the rigid, interpretable, and statistically-driven SARIMA models to the flexible, non-linear, but opaque LSTM network. The choice of the best model depends not only on raw accuracy (Table 1) but also on the project's requirements for interpretability, complexity, and the nature of the data itself.

Future Work:

- **Hybrid Models:** A promising avenue is to create a formal **ARIMA-LSTM** hybrid model, where ARIMA models the linear component and an LSTM is trained on the non-linear residuals.
- **Multivariate Analysis:** If other relevant variables are available, this project could be extended to a multivariate forecast using models like **VAR** (Vector Autoregression) or a **multivariate LSTM**.
- **Probabilistic Forecasting:** All models in this project produced point forecasts. A more advanced approach would be to generate probabilistic forecasts (prediction intervals), which quantify the uncertainty of the forecast.